ARPAN SHINGALA



CONTACT

Phone: +49(0)1523 1812206
E-Mail: av.shingala02@outlook.com
LinkedIn: linkedin.com/in/arpan-shingala02

Rintheimer Hauptstraße 70,
76131 Karlsruhe

Work authorisation: valid student visa, eligible for employment in Germany

EDUCATION

Master of Science (M.Sc.) Computer Science 10/2026 – expected 10/2028
Karlsruhe Institute of Technology | Karlsruhe

Bachelor of Technology (B.Tech) Mathematics & Computing 09/2020 – 05/2024
Dhirubhai Ambani University | Gandhinagar, India

- **Relevant coursework:** Data Structures & Algorithms, OOP, DBMS, Linear Algebra, Mathematical Statistics, Mathematical Optimisation, Machine Learning
- **Bachelor's thesis:** "Learning-Based Fuzzing of QUIC implementations" (DRDO-sponsored research)
- **Final grade:** 8,68/10 (≈ German grade 1,5)

WORK EXPERIENCE

Software Developer 11/2025 – 08/2026
Silicon Bits | Ahmedabad, India

- Refined image-processing models that detect known and unknown logos and extract predefined invoice details using low-resource techniques, deployed as a local pipeline accessible through APIs and a user interface.
- Developed automation systems for real-time applications by programming PLCs and provided support for an industrial client of Silicon Bits.
- Developed a real-time laser-based distance measurement system using centroid detection and triangulation, improving processing speed from 3 to 30 FPS through algorithm design and hardware utilisation for industrial use.

Freelancer 01/2025 – 10/2025
Self-Employed | Remote, India

- Designed and built a secure trading platform with authentication, encryption and real-time monitoring, including protections against common cyberattacks.
- Worked directly with clients to refine features and meet their requirements while improving scalability.

Junior Python Developer 10/2024 – 12/2024
Asquare Tech Lab | Rajkot, India

- Improved a Punjabi acoustic model by 32% in accuracy by integrating new data in real time into the SRT subtitle workflow.
- Built a UI tool that calculates transcription accuracy automatically and fixed line break handling for consistent SRT output.

Research Assistant Sponsored by DRDO 01/2024 – 04/2024
Dhirubhai Ambani University | Gandhinagar, India

- Built an automated, state-machine-guided fuzzing tool for gQUIC (used the L* algorithm, modified quic-scapy for gQUIC 43) that generates test sequences to detect bugs in lsquic.
- Validated the tool by reproducing a known critical vulnerability and identified previously undocumented behaviour in the lsquic library.

Quality Assurance Intern 05/2023 – 07/2023
Simform Solutions | Ahmedabad, India

- Assessed two internal web applications for OWASP Top 10 vulnerabilities using Nmap, Acunetix, Retire.js and Shodan, and practised reflective XSS exploitation on PortSwigger labs.

Research Assistant Sponsored by DRDO 01/2023 – 05/2023
Dhirubhai Ambani University | Gandhinagar, India

- Assessed methodologies for analysing QUIC protocol implementations, focusing on identifying vulnerabilities and performance bottlenecks. Proposed and validated potential techniques for bug detection in QUIC protocols.

SCHOOL

Higher Secondary School Certificate (GHSEB) 03/2020

Dev International School | Junagadh, India

- **Final grade:** 81,5/100 (≈ German grade 1,8)

UNIVERSITY PROJECT

Learning state machines of QUIC implementations (Semester 7) 07/2023 – 11/2023

- Explored state machine modelling as a systematic approach to simplify and analyse the complex protocol flows of QUIC, specifically focusing on automating state machine generation using the L* algorithm.
- Modified existing tool (quic-scapy) to improve packet processing for a newer version of QUIC implementation (gQUIC 43).

Replay attack detection using BSIF (Semester 7) 07/2023 – 11/2023

- Designed and implemented a hybrid image spoof detection model using BSIF filters and YC_rC_b, L*u*v* colour spaces, enhancing accuracy and robustness in detecting live v/s replayed faces.
- Conducted extensive testing to validate the model's resilience against replay attacks, ensuring consistent performance and security across various devices.

Static Analysis to Identify Security Vulnerabilities in Code (Semester 7) 07/2023 – 11/2023

- Developed a static analysis tool using Abstract Syntax Trees (ASTs) to identify SQL injection vulnerabilities, detecting insecure patterns in Python code.
- Analysed query patterns for function calls **execute** and **executemany**, flagging non-prepared statements as security risks.

TECHNICAL SKILLS

Programming languages

Python	very good knowledge
C/C++	good knowledge
Rust	good knowledge
JavaScript / TypeScript	good knowledge

Security and networking

Wireshark, Nmap	good knowledge
Protocol fuzzing, state-machine learning	good knowledge
Static analysis (AST-based)	good knowledge

Machine learning and data

scikit-learn, TensorFlow, NumPy, Pandas	very good knowledge
PostgreSQL, MongoDB	good knowledge

Web, tools and systems

React / Next.js / Node.js	good knowledge
Linux, Bash, Git	very good knowledge

LANGUAGES & INTERESTS

Languages

German	basic knowledge (A1, in progress)
English	very good knowledge (IELTS 7.5)
Hindi	very good knowledge
Gujarati	mother tongue

Interests

Cycling, reading, movies

VOLUNTARY WORK

Microsoft Student Technical Club 10/2021 – 07/2023

- Promoted awareness of open-source technologies through club activities and initiatives.
- Organised and facilitated technical workshops and hackathons, fostering a collaborative learning environment.

Karlsruhe, 01.11.2026

